

THE PASCH-SWITCH QUOTIENT GRAPH OF STS(15) HAS DIAMETER 11

ZIJIAN ZENG

ABSTRACT. There are 80 isomorphism classes of Steiner triple systems of order 15. Prior work established that Pasch switching gives one component on 79 classes and one isolated anti-Pasch class, and published a complete per-switch table, but we did not locate a stated diameter for the nontrivial component. We reconstruct the simple quotient graph from canonical representatives and certify that it has 258 edges and component diameters 11 and 0. Exactly two unordered pairs attain distance 11 in our canonical numbering. The certificate contains all 80 representatives, all 1,390 Pasch occurrences, the complete edge list, a geodesic and BFS layers, and a fail-closed verifier. The verifier uses a second Pasch enumerator and exact Steiner-quasigroup isomorphisms, without nauty or discovery output. Exhaustion of the vertex universe relies on the classical 80-class classification and is cross-checked against the complete DesignTheory.org catalogue.

1. INTRODUCTION

A Steiner triple system of order v , or STS(v), is a 3-uniform set system in which every pair of points lies in exactly one block. The classification at order 15 has 80 isomorphism classes [1, 7]. A Pasch configuration is a four-block Steiner trade on six points: its blocks may be replaced by the other four blocks with the same pair coverage.

Gibbons computed that 79 of the 80 classes are mutually reachable by 4-cycle (Pasch) switches. Grannell, Griggs, and Murphy independently verified that result and published the outcome of every 4-cycle switch in the standard numbering [4, 5]. Their quotient graph consists of a 79-vertex component and the unique 4-cycle-free class as an isolated vertex. The same prior boundary is summarized in later cycle-switching work [6]. The switch table determines the graph, but neither that paper nor the later sources checked in our literature audit state its diameter.

Our result is the following exact finite computation.

Theorem 1.1. *Let G be the undirected simple graph on the 80 isomorphism classes of STS(15), with two distinct classes adjacent when one Pasch switch maps a representative of one class to the other class. Then G has 258 edges and connected components of orders 79 and 1. The component diameters are respectively*

$$11 \quad \text{and} \quad 0.$$

The 79-vertex component has radius 6.

This is a certified finite result, not a new classification of STS(15). The contribution is the exact graph metric and a replayable modern certificate. The 1999 switch table is acknowledged as prior data sufficient in principle to derive the metric.

2. DEFINITIONS AND QUOTIENT CONVENTION

Fix $X = \{0, 1, \dots, 14\}$. A labelled STS(15) is a set $B \subseteq \binom{X}{3}$ such that

$$|\{A \in B : p \subseteq A\}| = 1 \quad \text{for every } p \in \binom{X}{2}.$$

Consequently $|B| = 35$. Two systems are isomorphic if a permutation of X maps one block set to the other.

For six distinct points, write a Pasch and its mate as

$$\begin{aligned} P &= \{abc, ade, fbd, fce\}, \\ P' &= \{abd, ace, fbc, fde\}. \end{aligned}$$

The two sides cover the same 12 pairs. Thus $B^P = (B \setminus P) \cup P'$ is again an STS(15). Relabeling commutes with this operation, so switching all Pasches in one representative finds every quotient neighbor of its class.

We use the *simple* quotient convention. A switch to an isomorphic copy of the source is a loop and is discarded. Several Pasches may lead to the same target class; these give one edge. A singleton component has diameter zero.

3. CONSTRUCTION AND EXACTIFICATION

3.1. Representatives. The discovery construction starts with the projective system $PG(3, 2)$: its points are the nonzero vectors of $\mathbf{F}_2^4$, and $\{x, y, x + y\}$ is a block. Closing this seed under all Pasch switches and canonicalizing after every switch produces 79 incidence-graph canonical forms. For the remaining class we use the Bose construction on $\mathbf{Z}_5 \times \mathbf{Z}_3$, with $x \circ y = (x + y)/2$ in $\mathbf{Z}_5$; direct enumeration shows that this system has no Pasch.

For naming only, a system is encoded as its bipartite incidence graph with 15 degree-7 point vertices and 35 degree-3 block vertices. We use `labelg` from nauty 2.9.3 to obtain canonical graphs; nauty's graph isomorphism algorithms are described in [8]. Extracting the degree-7 vertices and sorting the resulting triples gives identifiers V000, ..., V079. The decisive verifier does not trust these identifiers.

As a classification cross-check, the set of 80 canonical incidence graphs is exactly equal to the complete 80-record DesignTheory.org catalogue [2]. Its compressed source has SHA-256

11310c7e35330e842938dbcc4bb6e59144bf40576323f22499a809364d50f2c9.

3.2. Independent switch reconstruction. Discovery identifies a Pasch by inspecting each four-block subset. The verifier instead uses the following standard cycle description. For points $x \neq y$, remove x, y and the third point of their block. The blocks through x and through y induce two perfect matchings on the remaining 12 points. Their union is a disjoint union of even cycles; each 4-cycle gives a Pasch [5, 6]. Set deduplication accounts for the three opposite point pairs of the same Pasch.

The verifier finds 1,390 Pasch occurrences. For each, it independently constructs the other four triples having the same 12 covered pairs and checks that the switched object again covers all 105 pairs exactly once. Let $p_S(x)$ be the number of Pasches of S whose six-point support contains x . The invariant

$$(|\mathcal{P}(S)|, \text{sort}(p_S(0), \dots, p_S(14)))$$

has 80 distinct values on the released representatives. It therefore proves pairwise nonisomorphism and identifies a unique candidate target for every switch. The verifier then constructs a full block-preserving bijection to that target by backtracking on the Steiner quasigroup operation. No hash equality is treated as an isomorphism proof.

After discarding 83 self-switch occurrences and deduplicating parallel targets, this replay gives 258 undirected edges. Every edge has a switch witness from both directions.

4. DISTANCE CERTIFICATE

Table 1 records the main exact counts. All values are integers obtained from the serialized representatives and edge list.

TABLE 1. Certified quotient-graph data.

Quantity	Value
Isomorphism classes	80
Pasch occurrences over all representatives	1,390
Self-switch occurrences	83
Simple nonloop edges	258
Component orders	79, 1
Component diameters	11, 0
Radius of the 79-component	6
Unordered diameter pairs	2

In canonical numbering, the two pairs at distance 11 are

$$\{V000, V006\}, \quad \{V000, V028\}.$$

One shortest path is

$$\begin{aligned} V000 - V047 - V002 - V027 - V054 - V018 - V035 - V037 \\ - V066 - V014 - V023 - V006. \end{aligned}$$

It has 11 edges. BFS from V000 gives layer sizes

$$1, 1, 3, 9, 11, 9, 9, 8, 11, 12, 3, 2,$$

with V006 and V028 in the final layer. Hence their distance is at least 11, while the displayed path gives the reverse inequality. Independent BFS from every vertex of the 79-component has eccentricity at most 11, proving the global upper bound in Theorem 1.1.

5. CERTIFICATE, AUDIT, AND LIMITATIONS

The certificate comprises the 80 normalized representatives, the complete switch-occurrence table, the 258-edge list, all components and eccentricities, both diameter pairs, one geodesic, a complete distance row, and its BFS layers. The principal input digests are

representatives: 8a2044e2613d4210866cf635a82a0d91e95f0954f91c705e207e92736e970b23;
edge list: efcf41db2c383682b8ff586b610aae6d6130fb412d3e05191dd277fa2cab91af;
result: fdf6b0ab98823674456a40636089395d6eb54636ca0d0f6473dd3f03abc591c5.

The fail-closed verifier parses duplicate-free JSON, validates all Steiner systems, reconstructs all switches and exact target isomorphisms, compares the complete edge set, and reruns all-pairs BFS. A separate short BFS program verifies the graph metric from the edge file alone. Corruption tests confirm rejection of a malformed block, a missing vertex, a deleted edge, a duplicate edge, a false diameter, and a false switch target.

The proof is exact but computational. It does not reprove the historical classification that there are exactly 80 classes. Instead it verifies pairwise nonisomorphism of the released systems and binds their canonical set to the complete external catalogue. The novelty statement is likewise bounded: two searches, the second using the value 11 and the edge count 258, located no publication stating the diameter. Recent work on cycle-switching graphs at order 19 has a different endpoint [3].

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

All certificate data and Python source are supplied with this manuscript. The full verification command is

```
python3 code/independent_verifier.py certificate.
```

The graph-only check is

`python3 code/verify_bfs.py.`

Discovery used nauty 2.9.3 for canonical naming. Certification uses Python standard-library exact integer and set operations only. No floating-point calculation, random sampling, optimizer, or proof assistant was used.

OpenAI Codex was the sole computational research assistant. It was used to search the literature, implement the independent enumeration and verification programs, run adversarial tests, and prepare the manuscript. The named author selected the problem and is responsible for reviewing the mathematical claims and submission. No additional author is claimed.

CONFLICT OF INTEREST AND FUNDING

The author declares no conflict of interest. No external funding was used for this computation.

REFERENCES

1. F. N. Cole, Louise D. Cummings, and H. S. White, *The complete enumeration of triad systems in 15 elements*, Proceedings of the National Academy of Sciences of the United States of America **3** (1917), no. 3, 197–199.
2. DesignTheory.org, *Database of t-designs*, Queen Mary University of London, Accessed 30 August 2026.
3. Grahame Erskine and Terry S. Griggs, *Cycle switching in steiner triple systems of order 19*, Journal of Combinatorial Designs **33** (2025), no. 5, 195–204.
4. Peter B. Gibbons, *Computing techniques for the construction and analysis of block designs*, Ph.D. thesis, University of Toronto, 1976, Technical Report 92.
5. M. J. Grannell, T. S. Griggs, and J. P. Murphy, *Switching cycles in steiner triple systems*, Utilitas Mathematica **56** (1999), 3–21.
6. Petteri Kaski, Veli Mäkinen, and Patric R. J. Östergård, *The cycle switching graph of the steiner triple systems of order 19 is connected*, Graphs and Combinatorics **27** (2011), no. 4, 539–546.
7. R. A. Mathon, K. T. Phelps, and A. Rosa, *Small steiner triple systems and their properties*, Ars Combinatoria **15** (1983), 3–110, Errata: Ars Combinatoria 16 (1983), 286.
8. Brendan D. McKay and Adolfo Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation **60** (2014), 94–112.

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: zijianzeng@foxmail.com

Email address: 1002266693@ucsiuniversity.edu.my